KHÁNH TRÌNH – MAI ANH – GIA AN – MINH TÚ – MINH KHANG

FRIDGI

FRIDGI

PROBLEMS

+ 60% of food waste happens at home (UNEP)
+ ~30% of fresh fruit is thrown away at home (WRAP)
+ $1,500–$1,800 lost per family yearly (ReFED & NRDC)

Key problem: Lack of clear, real-time indicators of when fruit is edible leads to unnecessary household waste.

SOLUTION

ACTIVE **SMART** **CONTAINER**

Automates food storage management

FRIDGI

→ Tracks air changes in container

→ Alert users (via smartphones)

→ Slow down rotting

PRODUCT

ACTIVE INTERVENTION

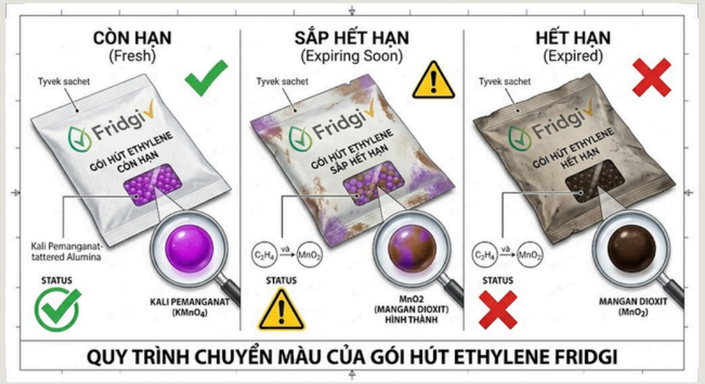
If consumption is delayed, the container turns on an internal micro-fan to push air through a potassium permanganate filter pad, removing aging gases and extending shelf-life by five to seven days.

HARDWARE

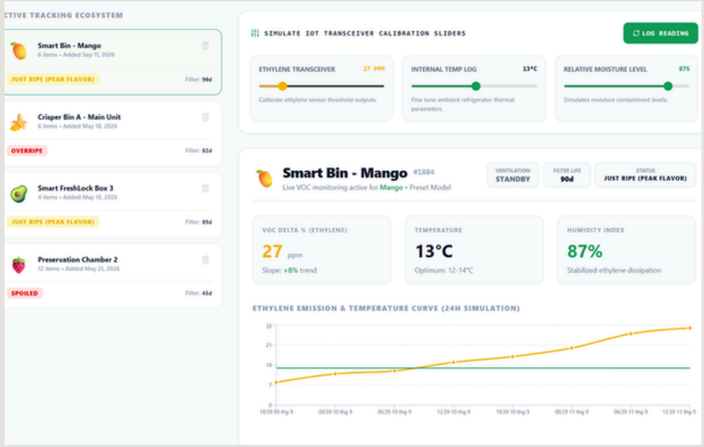
Features a clear Tritan container body paired with an airtight lid holding an ESP32 microchip, a calibrated gas sensor, a humidity reader, and a small fan system.

SOFTWARE

Tracks the rate of gas change over time based on the specific fruit type selected within the interface. The system monitors the rising data line and sends a phone notification when perfect ripeness occurs.

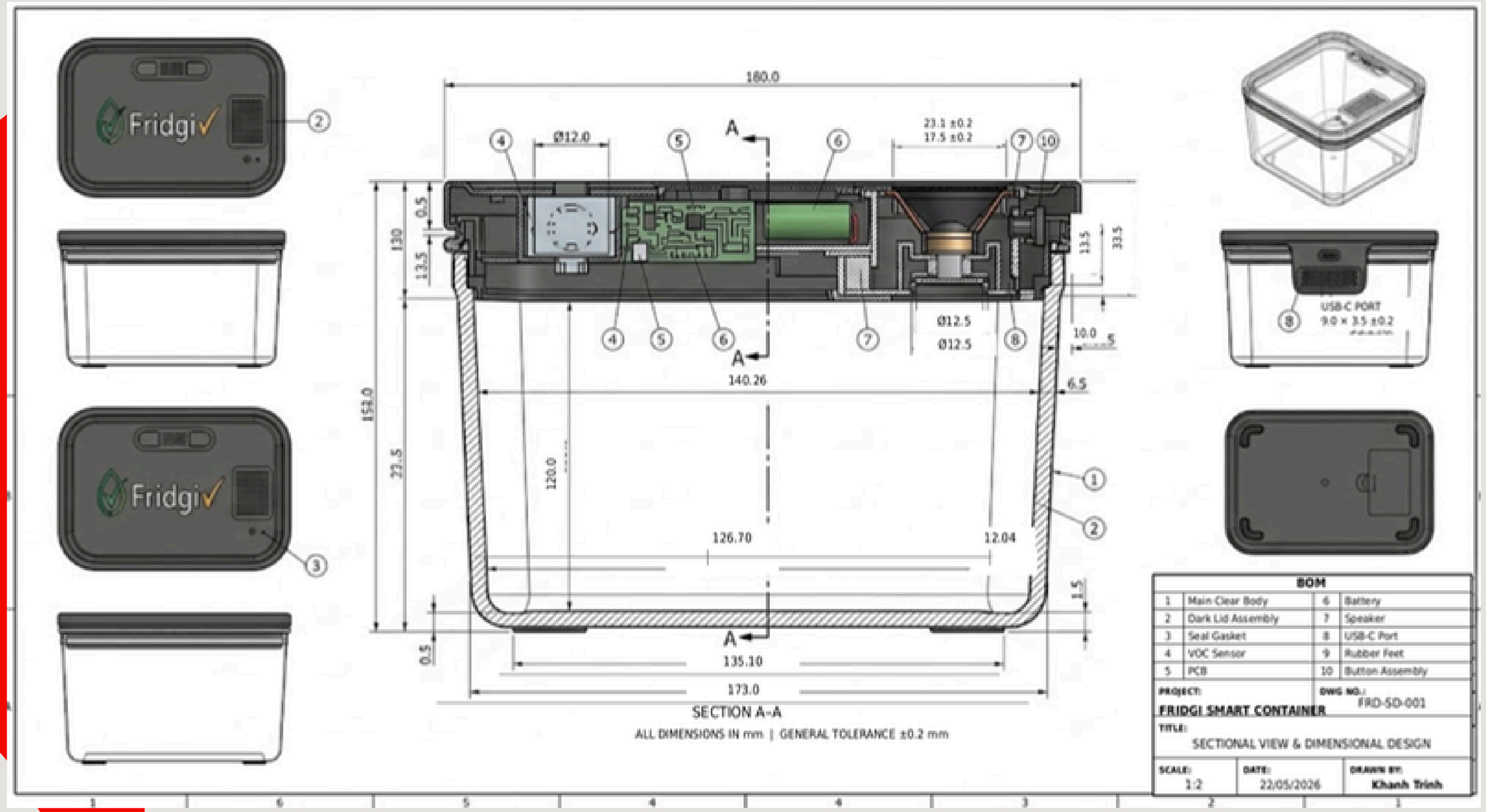


Apps QR



https://fridgi-407334579894.asia-southeast1.run.app

BLUEPRINT


Fridgi
Fridgi
180.0
Ø12.0
23.1 ±0.2
17.5 ±0.2
130
153.0
22.5
0.5
13.5
33.5
10.0
5
6.5
1.5
Ø12.5
Ø12.5
140.26
120.0
126.70
12.04
135.10
173.0
SECTION A–A
ALL DIMENSIONS IN mm | GENERAL TOLERANCE ±0.2 mm
USB-C PORT
9.0 × 3.5 ±0.2
BOM
1 Main Clear Body
2 Dark Lid Assembly
3 Seal Gasket
4 VOC Sensor
5 PCB
6 Battery
7 Speaker
8 USB-C Port
9 Rubber Feet
10 Button Assembly
PROJECT:
FRIDGI SMART CONTAINER
DWG NO.:
FRD-SD-001
TITLE:
SECTIONAL VIEW & DIMENSIONAL DESIGN
SCALE:
1:2
DATE:
22/05/2026
DRAWN BY:
Khanh Trinh

CUSTOMER SEGMENT

B2C: SMART & ECO-FRIENDLY HOMES.

- ***Who***: Mid-to-high-income city families, busy professionals, and healthy eating enthusiasts.
- ***The Problem***: They waste money by throwing away up to 30% of their fresh fruit because it rots unseen inside deep fridge drawers.
- ***How We Make Money***: They buy the Smart Container once, then regularly buy replacement Tyvek filter packs every 90 days (Razor-and-Blade model).

B2B: SUPERMARKETS & GROCERY STORES

- ***Who***: Store managers and supply chain directors for organic shops and large supermarket chains.
- ***The Problem***: They lose a lot of profit because 10% to 15% of their fresh fruit spoils on open shelves before customers can buy it.
- ***How We Make Money***: Bulk sales of commercial shelf hardware + monthly fees for our smart data software (SaaS) that helps them track inventory.

DIFFERENCE

AUTOMATION VS MANUTAL SETTINGS

COMPETITIORS (like TupperWare or OXO)

Rely on manual plastic vents
→ require constant guesswork

FRIDGI

Operates on automatic, real-time environmental data

LOWER COSTS THROUGH SOFTWARE

COMPETITIORS (like Industrial Gas Analyzers)

Industrial gas tracking equipment
→ costs thousands of dollars

FRIDGI

- Affordable sensors
- Increase accuracy: advanced algorithm filtering environmental noise & fruit volume variations

FINANCIAL

Cost of Goods Sold (BOM): 100,000 VND per unit at mass manufacturing scale.

Pricing Strategy:

+ Standard Edition (Tracking and application alerts): 399,000 VND.
+ Premium Edition (Integrated active filtration system): 549,000 VND.
+ Profitability: Upfront hardware sales yield a consistent profit margin of fifty to sixty percent.

BUSINESS MODEL

INDIVIDUAL CUSTOMERS (B2C)

HARDWARE SALES

- Upfront revenue comes from selling smart containers
- *Standard Edition*: 399,000 VND → basic tracking
- *Premium Edition*: 549,000 VND → tracking + active air filtering

FILTER REPLACEMENTS

- *Chemical filters: stop working after three months → be replaced*
- *Replacement packs: 49,000 VND → continuous sales from the existing customer base*

SUPERMARKET SOLUTIONS (B2B)

SHELF KIT INSTALLATIONS

- Smart sensor strips & large filter modules → sold to supermarkets & warehouses (pricing based on total shelf length)

SOFTWARE SUBSCRIPTIONS

- Supermarkets: monthly subscription fee of 1,200,000 VND/store → a cloud dashboard that monitors fruit freshness in real time and reduces food waste

BULK FILTER SALES

- Large supermarket replacement filters → supplied through annual contracts → stable recurring revenue

BUSINESS MODEL

Future Expansion

TECHNOLOGY LICENSING

Long-term growth relies on selling the rights to the unique fruit data and software algorithms directly to global refrigerator manufacturers for a royalty fee per unit.

MARKETING STRATEGIES

VISUAL PROOF CAMPAIGNS

Visual proofs using time-lapse comparisons to showcase freshness and cost savings

NICHE COMMUNITY INTEGRATION

Partnerships with fitness communities to captivate people’s attention

CROWDFUNDING LAUNCH

Crowdfunding launch to establish trust and support initial production setup

MARKET PLAN

PHASE 1 – LOCAL TESTING

Deploy a one-hundred-unit pilot test to tech-focused families and partner with a local organic grocery store to improve algorithm accuracy under real retail conditions.

PHASE 2 – SUBSCRIPTION PARTNERSHIP

Partner with major regional supermarket chains to install B2B kits on high-value fruit displays while selling B2C containers at checkout lanes.

PHASE 3 – DIGITAL & ENTERPRISE SCALING

Establish verified storefronts across major regional online sales platforms while expanding B2B operations to large cold chain logistics warehouses.

PHASE 4 – APPLIANCE INTEGRATION & LICENSING

Partner with global appliance brands to integrate gas-tracking technology into next-generation household refrigerators. This expands the business into high-margin software licensing and royalty revenue per refrigerator sold, while also providing food exporters with predictive freshness analytics.

MILESTONES

YEAR 1: MONTH 1–3

Finalize mold designs and high-humidity sensor testing for containers and shelf strips

MONTH 4–6

Launch crowdfunding campaign, secure local manufacturing partners, and begin closed mobile app trials

MONTH 8–12

Start first B2C mass production, fulfill pre-orders, and launch enterprise supermarket dashboard platform

SWOT ANALYSIS

STRENGTH

Strong hardware profit margins paired with regular filter revenue and B2B software subscriptions; smart data processing software that lowers physical production costs.

WEAKNESS

Requires manual user input in the mobile app to select the fruit type before tracking can begin.

OPPORTUNITIES

Rapidly expanding global market demand for smart-home automation and eco-friendly retail technologies designed to cut food supply chain waste.

THREAT

Rapid copycat manufacturing from older kitchenware brands, protected by getting patents early on the data-filtering software logic rather than the generic plastic container shape.

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PROTOTYPE

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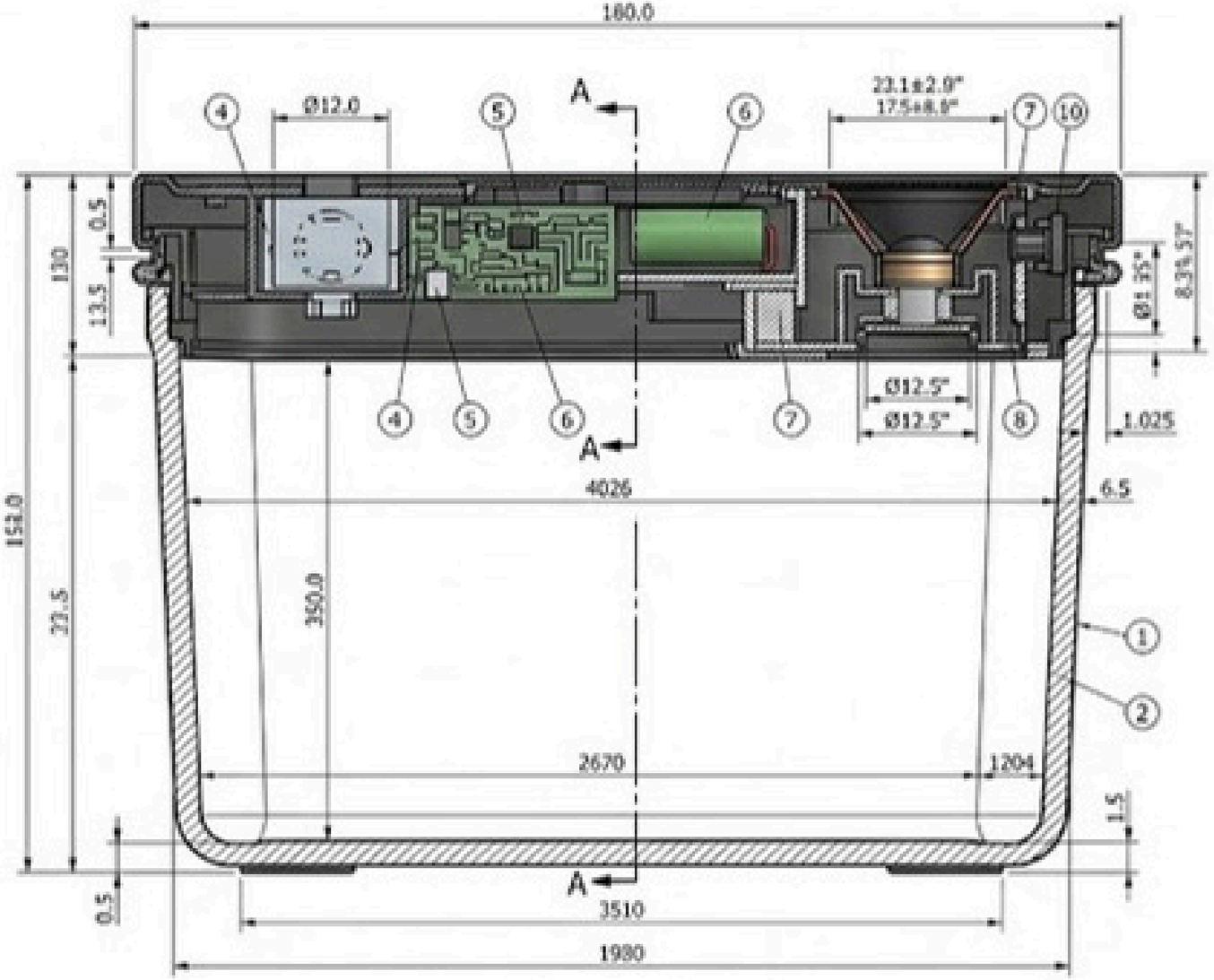
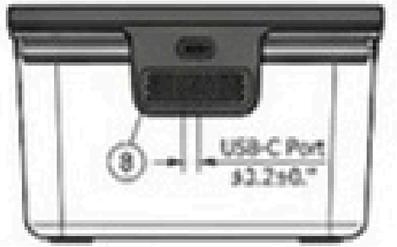
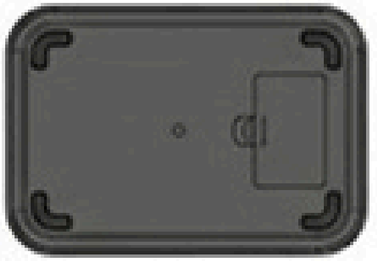


Ký hiệu mặt cắt (A-A)





BOM			
1	Main Clear Body	6	Battery
2	Complex Dark Lid Assy	7	Speaker
3	Seal Gasket	8	USB-C Port
4	VOC Sensor	9	Rubber Feet
5	PCB	10	Button Assy
Pos.	Material	BOM	Marterio

PROJECT: FRIDGI SMART CONTAINER	DWG NO.: FRD-SD-001
TITLE: SECTIONAL VIEW & DIMENSIONAL DESIGN	
SCALE: 1:2	DATE: 22/05/2026
DRAWN BY: ENGINEER	

CÒN HẠN
(Fresh)
Tyvek sachet
Fridgi
GÓI HÚT ETHYLENE
CÒN HẠN
Kali Pemanganat-
tattered Alumina
STATUS
KALI PEMANGANAT
(KMnO4)
SẮP HẾT HẠN
(Expiring Soon)
Tyvek sachet
Fridgi
GÓI HÚT ETHYLENE
SẮP HẾT HẠN
C2H4
và
MnO2
STATUS
MnO2
(MANGAN DIOXIT)
HÌNH THÀNH
HẾT HẠN
(Expired)
Tyvek sachet
Fridgi
GÓI HÚT ETHYLENE
HẾT HẠN
C2H4
và
MnO2
STATUS
MANGAN DIOXIT
(MnO2)
QUY TRÌNH CHUYỂN MÀU CỦA GÓI HÚT ETHYLENE FRIDGI

Fridgi.

HOME HUB (B2C) 3 | RETAIL (B2B) 4

SAVINGS OPTIMIZED
$148.50

OK

GEOMETRIC EDGE-CORE PRESERVATION

B2C Home Crisper - Keep Your Produce Fresh for 7 Days More

Active ventilation control systems automatically filter ethylene and match moisture settings for fruit types to maximize home health. Scan barcodes or select presets below to begin.

ACTIVE SENSORS
9 Channels

ECOSYSTEM STATE
BALANCED

Fruit Bio-Profiling Node Editor

Configure simulated IoT telemetry triggers for active preservation models

1. TAP TO SELECT ACTIVE FRUIT MATRIX MODEL

Banana | Avocado | Strawberry | Mango | Peach | Papaya

CONTAINER / BOX LABEL

e.g., Left Crisper drawer

QUANTITY (UNITS)

6

SIMULATED ETHYLENE (PPM) 25 ppm

START TEMPERATURE INDEX 14°C

START RELATIVE HUMIDITY 88%

LAUNCH HOME PRESERVATION BIO-SENSOR MODULE

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HOME HUB (B2C) 3 | RETAIL (B2B) 4

SAVINGS OPTIMIZED
$148.50

OK

GEOMETRIC EDGE-CORE PRESERVATION

B2B Commercial Dashboard - Reduce Food Shrink and Maximize Margins

Continuous IoT sensor monitoring and automated potassium permanganate filters keep perishable inventory at peak status. Leverage dynamic price markdowns to optimize velocity before decay thresholds occur.

ACTIVE SENSORS
12 Channels

ECOSYSTEM STATE
BALANCED

Bulk Shipment Registry Interface

Configure simulated IoT telemetry triggers for active preservation models

FRUIT SHIPMENT CATEGORY
Banana Model

TARGET RETAIL SHELF MAPPING
Commercial Display Wall A2 (Eco)

BATCH TOTAL WEIGHT (KG)
100

BASE RETAIL PRICE ($ / KG)
$ 3.49

SHIPMENT EAN BARCODE MAPPING
5011029348572

SCAN

GAS REVALUE 40 ppm

AMBIENT COMPARTMENT TEMP 15°C

AMBIENT COMPARTMENT HUMIDITY 87%

MAP LOT BATCH INTO COMMERCIAL INVENTORY GRID

Fridgi.

HOME HUB (B2C) 3 | RETAIL (B2B) 4

SAVINGS OPTIMIZED
$148.58

OK

POTASSIUM PERMANGANATE FAN PRESERVATION

Adds 5 to 7 days of shelf life by filtering ambient decayed ethylene

Fan State: ACTIVE VENTILATION RUNNING

Automatic Activation Threshold: **45 ppm**

Force Idle (Stop Exhaust Fans)

MOBILE PUSH-NOTIF STREAM SIMULATOR

"Spoilage hazard: Clean out container to prevent ethylene contamination."

Mobile Alert Protocol:
Automatically triggers direct low-friction banner notifications to the retail store manager's dashboard tablet.

B2B DYNAMIC PRICING OPTIMIZER

Automatic retail system tracking decaying stock and computing active consumer markdowns to flush inventories before spoilage threshold models trigger.

SUGGESTED DISCOUNT DISPATCH	ORIGINAL RETAIL PRICE	ADJUSTED SALE PRICING
30% Price Cut	$3.99 / kg	$2.79 / kg

$143.64

MARGINAL LOSS SAVED IF SOLD NOW

Publish Markdown to Labels

WEEKLY INVENTORY SHRINK	TOTAL SAVINGS THIS MONTH	STOCK ROTATION VELOCITY
1.2%	$1,840.00	+18.5%
Commercial Benchmark: 5.6%	Across active shelves A1-D6	Faster dispatch on overripe early alerts

AUTOMATED POTASSIUM PERMANGANATE PAD MARKETPLACE

Potassium permanganate pads saturate ethylene decay gas, degrading over 90 days.

One-Click Re-Order ($12.50)

Fridgi.

HOME HUB (B2C) 3 | RETAIL (B2B) 4

SAVINGS OPTIMIZED
$148.58

OK

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